



PYTHON ESSENTIALS COURSE SUBJECT

MOVIE TICKET BOOKING SYSTEM (CLI)

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PROJECT REPOSITORY

```
cd movie-ticket-booking
```

<https://github.com/lakebolt/movie-ticket-booking.git>

INTRODUCTION

The entertainment industry relies heavily on computerized reservation systems to streamline ticket distribution, monitor auditorium capacity, and enhance customer experience. Traditional paper-based ticket issuing or manual counter management often causes bottlenecks, scheduling errors, and duplicate seat allocations.

This project, Movie Ticket Booking System (CLI), is an automated, terminal-based software solution designed using Python. Built with a modular architecture, the system enables cinema patrons to view movie schedules, inspect real-time seating availability on a dynamic visual matrix, reserve multiple seats with automatic tax and invoice calculation, and perform hassle-free cancellations using a unique booking reference code. All data changes are persisted in a local JSON storage file to ensure data durability across application sessions.

PROBLEM STATEMENT

Manual or non-centralized cinema ticketing systems suffer from multiple operational vulnerabilities:

- **Duplicate Bookings:** Lack of real-time seat lock mechanisms can cause two customers to be assigned the same seat.
- **Human Billing Errors:** Calculating taxes, subtotals, and refund deductions manually is prone to arithmetic mistakes.
- **Lack of Visual Feedback:** Customers cannot easily visualize where they will sit relative to the cinema screen.
- **Session Data Loss:** Simple educational scripts store data only in RAM, losing all reservation histories once the program terminates.

Objective: To engineer a reliable, modular, command-line Python application that manages real-time seat matrices, computes itemized invoices with exact tax rates, enables booking cancellations with automated seat release, and stores cinema records persistently using standard file input/output.

FUNCTIONAL REQUIREMENTS

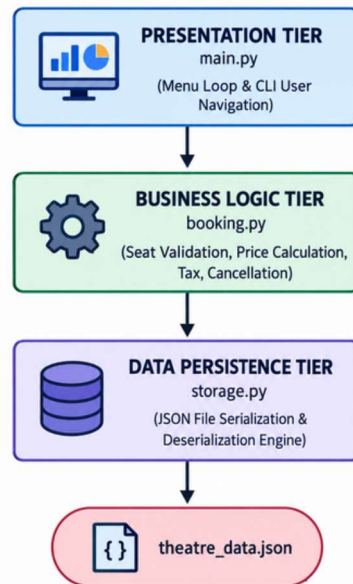
ID	Requirement Name	Description
FR-01	Movie Catalog Display	System must list all available movies with Movie ID, Title, Genre, Duration, and Ticket Price in INR (₹).
FR-02	Visual Seat Layout Matrix	System must generate a 2D seating grid indicating Available seats as [O] and Occupied seats as [X] along with screen orientation.
FR-03	Multi-Seat Reservation	System must allow booking one or more seats simultaneously (e.g., A1, A2) in a single transaction.
FR-04	Input & Boundary Validation	System must validate row letters, column numbers, and reject duplicate or already occupied seats.
FR-05	Automated Invoice & Tax Calculation	System must compute the subtotal, apply a 5% entertainment tax, and display a formatted confirmation slip.
FR-06	Unique Booking ID Generation	System must generate a randomized unique reference code (e.g., TKT-89K2M4) for tracking each reservation.
FR-07	Booking Cancellation	System must allow users to cancel reservations by reference code, calculating refund amounts and freeing seats.
FR-08	Ticket Lookup	System must retrieve customer details, booked seats, and payment information using the Booking ID.
FR-09	Persistent Storage	System must automatically save and load all changes to/from a local JSON database file (theatre_data.json).

NON- FUNCTIONAL REQUIREMENTS

- Modularity: The codebase must be cleanly partitioned into separate modules (storage.py, display.py, booking.py, and main.py) adhering to the separation of concerns.
- Portability: Must operate seamlessly across Windows, macOS, and Linux without platform-specific dependencies.
- Zero Third-Party Dependencies: Built strictly using Python's standard libraries (json, random, datetime, os, sys), ensuring zero installation friction on evaluation environments.
- Performance & Responsiveness: Terminal interactions, seat rendering, and data queries must execute within sub-100 millisecond response times.
- Robustness & Fault Tolerance: Must gracefully handle invalid numeric inputs, out-of-range choices, and file I/O errors without unexpected program termination.

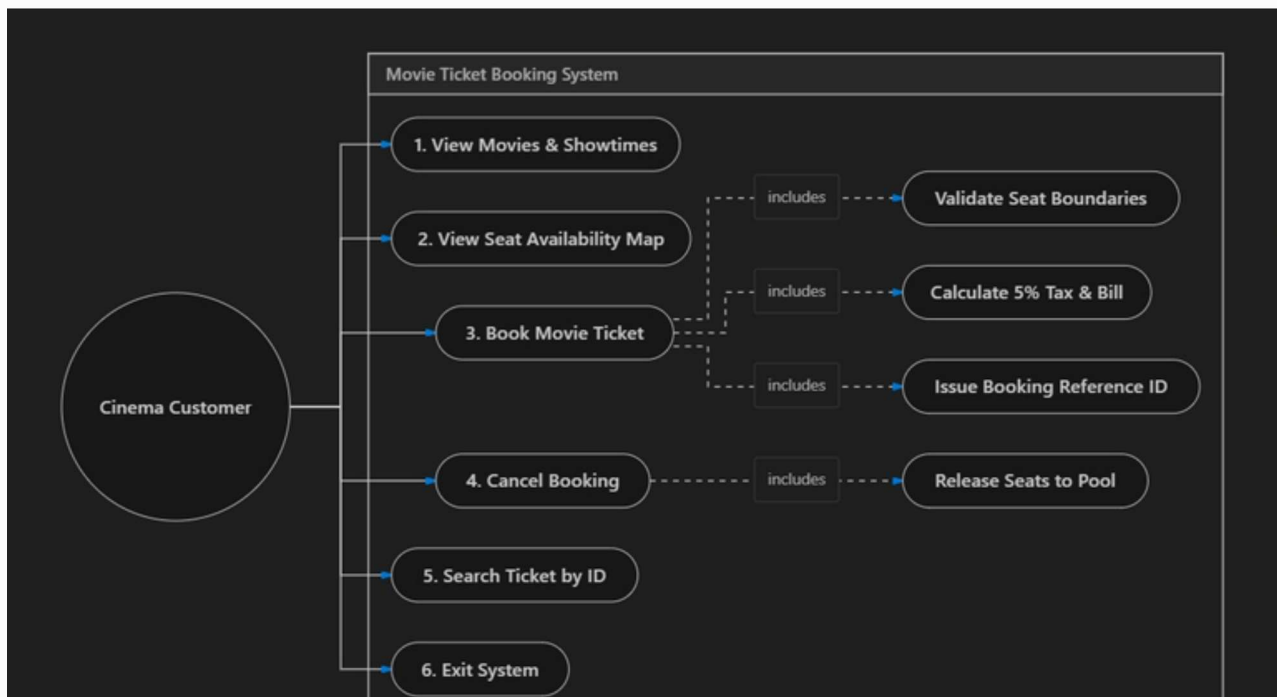
SYSTEM ARCHITECTURE

The application follows a 3-Tier Modular Architecture:

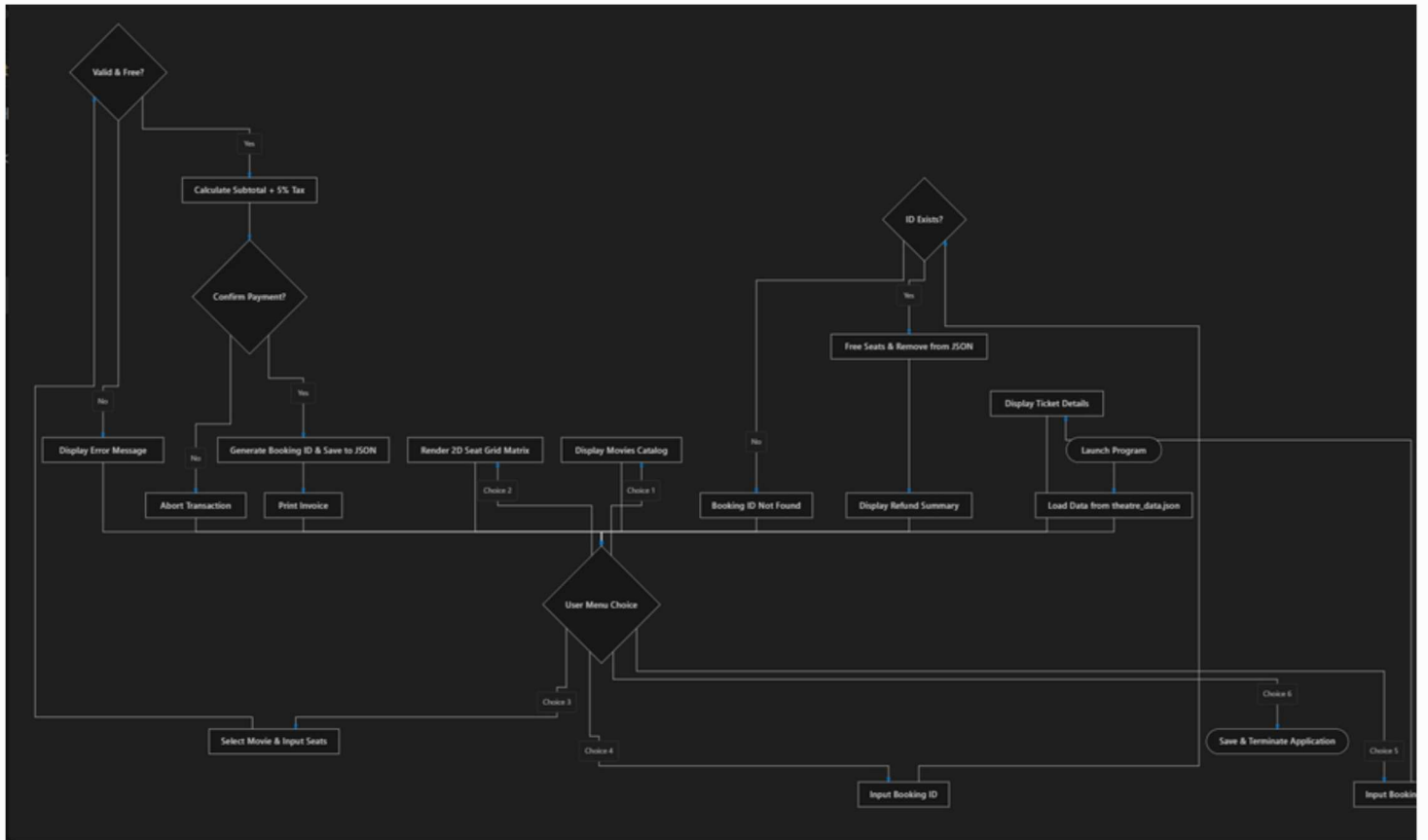


6.) DESIGN DIAGRAMS

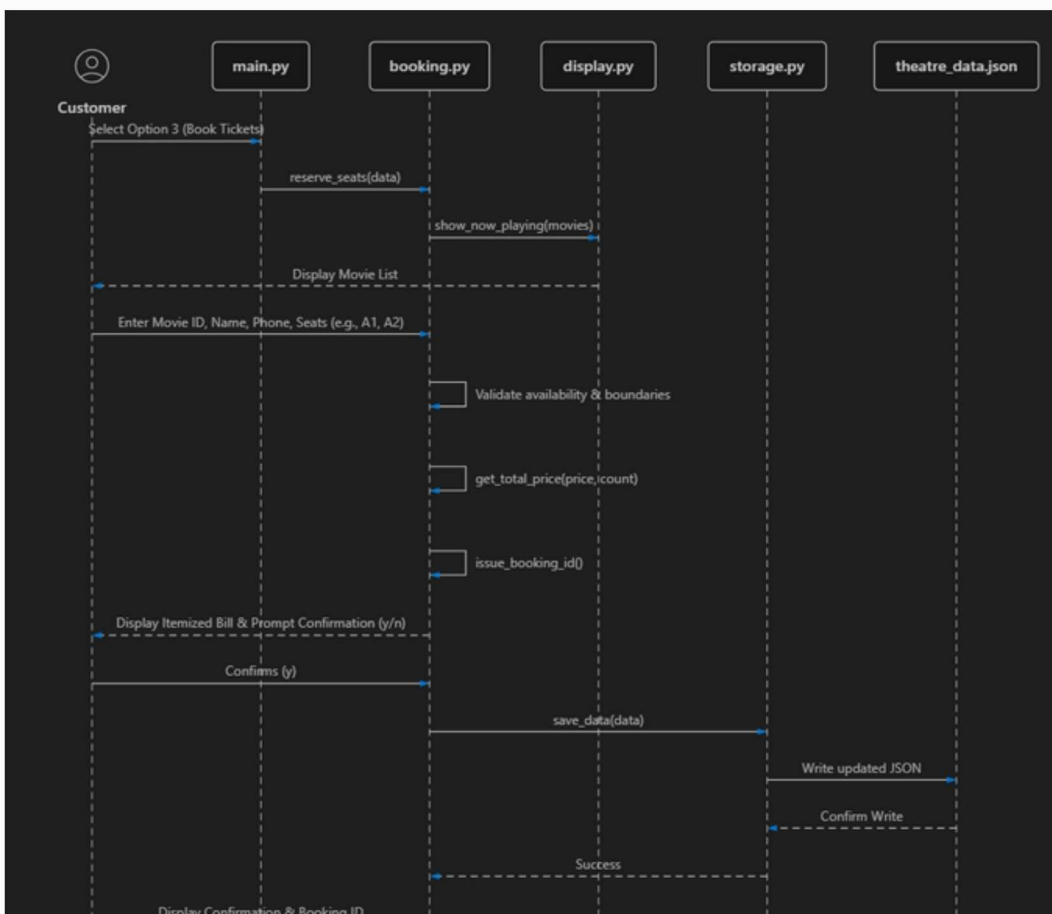
6.1 Use Case Diagram



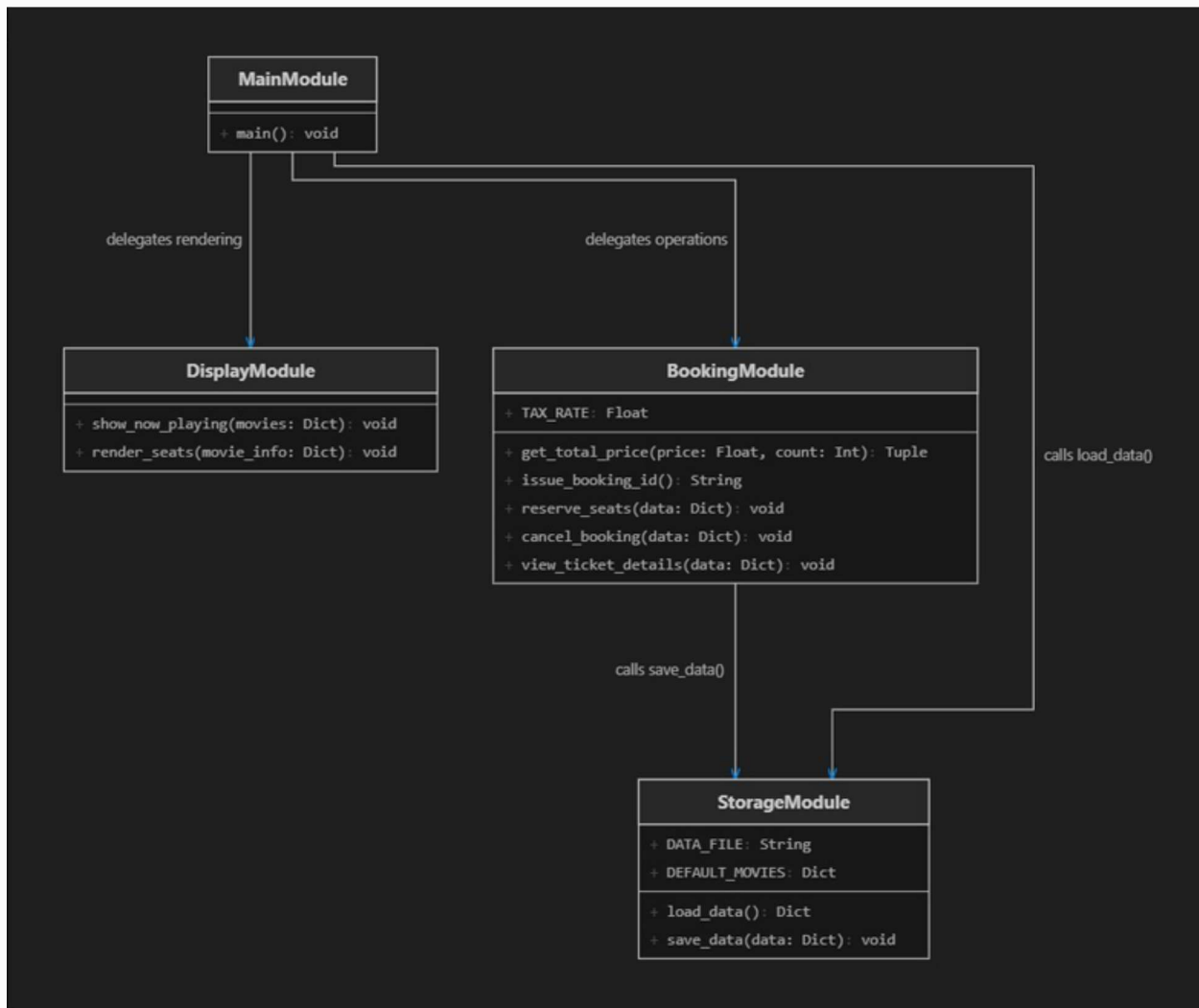
6.2 Workflow Diagram (Activity Flow)



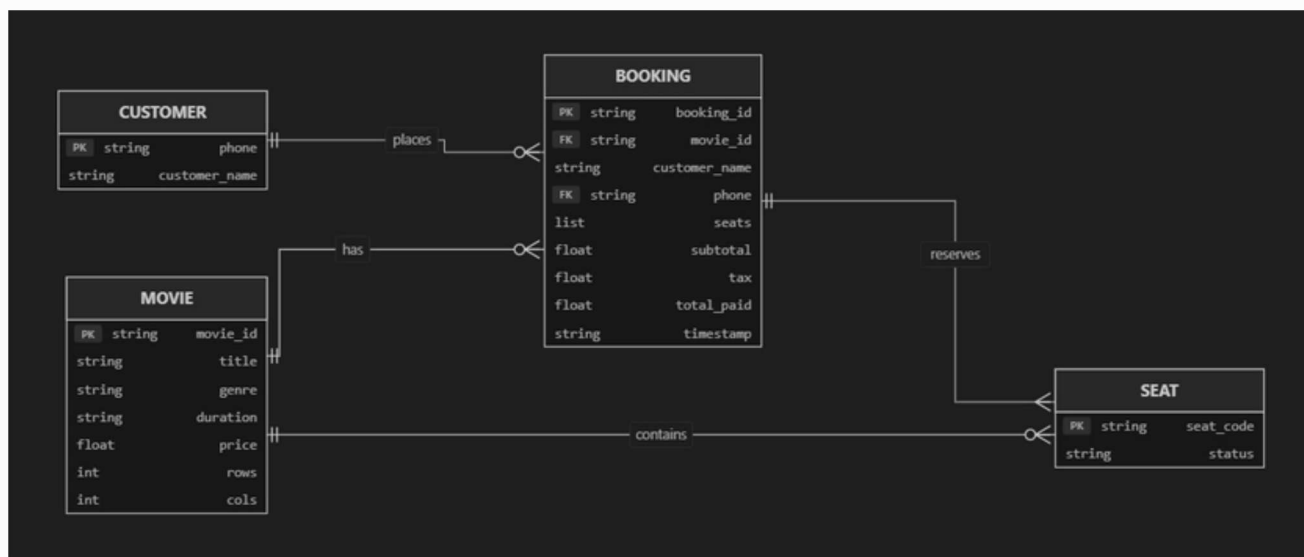
6.3 Sequence Diagram (Booking Transaction)



6.4 Class / Component Diagram



6.5 Entity-Relationship (ER) Schema



7.) . Design Decisions & Rationale

- Modular 3-File Separation:
 - Decision: Splitting functionality into storage.py, display.py, and booking.py.
 - Rationale: Isolates data access from presentation and business logic, satisfying the course modularity criterion and making the application easily unit-testable.
- Standard Library Only (zero external pip packages):
 - Decision: Using built-in modules (json, os, sys, random, datetime).
 - Rationale: Eliminates environment setup discrepancies. Evaluators can clone and run the program instantly without dependency installation errors.
- JSON-Based File Persistence:
 - Decision: Using theatre_data.json instead of SQLite or binary files.
 - Rationale: Human-readable, native Python dictionary serialization via json.load() and json.dump(), and easy verification of database state during evaluation.
- 2D Seat Grid Matrix:
 - Decision: Visual ASCII rendering with [O] and [X].
 - Rationale: Simulates a physical cinema screen layout directly within the terminal, improving user experience without violating the strict "no-GUI" constraint.

8.) Implementation Details

- storage.py:
 - Implements load_data() with fallback default initialization if the JSON file is missing or corrupted.
 - Implements save_data() with UTF-8 encoding and indented JSON formatting for clean readability.
- display.py:
 - show_now_playing(): Utilizes Python f-string format specifiers (e.g., f"{info['title']:<26}") to generate perfectly aligned tabular output.
 - render_seats(): Iterates through alphabet-indexed rows (chr(ord('A') + r)) and numeric columns to render the visual 2D matrix.
- booking.py:
 - get_total_price(): Implements tax logic $\text{total} = \text{subtotal} + (\text{subtotal} * \text{TAX_RATE})$ rounded to 2 decimal places.
 - issue_booking_id(): Combines random uppercase alphanumeric characters with a TKT- prefix.
 - reserve_seats() & cancel_booking(): Modifies state dictionaries and invokes save_data() for immediate persistence.
- main.py:
 - Controls user session loop (while True) and routes choices to their respective modular handlers.

9.) Screenshots / Execution Results

```
Problems 1 Output Debug Console Terminal Ports
Enter choice (1-6): C:\Users\laksh\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/laksh/Desktop/CSE ASS/main.py/VITyarthi_project/project.py"
[!] Invalid option. Please enter a number between 1 and 6.

#####
  🎬 MOVIE TICKET BOOKING SYSTEM 🎬
#####
1. View Movies & Showtimes
2. Check Seat Availability Map
3. Book Movie Ticket(s)
4. Cancel a Booking
5. Search Ticket by Booking ID
6. Exit
#####
Enter choice (1-6):
```

```
Enter choice (1-6): 2

=====
ID  Movie Title      Genre           Duration  Price
=====
[1] MARVEL: IRON MAN   Sci-Fi/Action   2h 6m     ₹300.00
[2] FORD V FERRARI    Sport/Action    2h 32m    ₹290.00
[3] MISSION IMPOSSIBLE Action/Thriller  1h 50m    ₹250.00
[4] THE PURSUIT OF HAPPYNESS Melodrama/Drama 1h 57m    ₹225.00
=====

Enter Movie ID to view seat map: 2

--- FORD V FERRARI Seat Map ---

----- SCREEN THIS WAY -----
   1  2  3  4  5
A [0] [0] [0] [0] [0]
B [0] [0] [0] [0] [0]
C [0] [0] [0] [0] [0]
D [0] [0] [0] [0] [0]
E [X] [X] [X] [0] [0]
F [0] [0] [0] [0] [0]
G [0] [0] [0] [0] [0]
H [0] [0] [0] [0] [0]

Legend: [0] = Available  [X] = Booked
Available Seats: 37/40
```

```
=====
                        BOOKING CONFIRMATION
=====
Booking ID   : TKT-89K2M4
Customer    : Rahul Sharma (9876543210)
Movie       : MARVEL: IRON MAN
Seats       : B1, B2 (Count: 2)
Subtotal    : ₹600.00
Tax (5%)    : ₹30.00
Total Amount : ₹630.00
Date & Time  : 2026-09-30 22:15:00
=====

[✓] Tickets booked successfully! Save your Booking ID.
```

10. Testing Approach & Test Cases

The application underwent rigorous manual validation against standard, boundary, and negative input scenarios:

Test ID	Test Scenario	Input Data	Expected Output	Status
TC-01	Display Movie List	Choice: 1	Catalog table with 4 movies in INR format (₹).	PASS
TC-02	Display Seat Grid	Choice: 2 , Movie ID: 1	6x7 grid rendering with legend and vacancy count.	PASS
TC-03	Valid Multi-Seat Booking	Seats: B1, B2	Subtotal, 5% tax calculated, invoice generated.	PASS
TC-04	Prevent Double Booking	Seats: A1 (Already booked)	System flags: [!] Seat(s) already occupied: A1 .	PASS
TC-05	Out of Bound Seats	Seats: Z9	System flags: [!] Invalid seat format/out of bounds .	PASS
TC-06	Invalid Phone Number	Contact: 123	System flags: [!] Please enter a valid phone number .	PASS
TC-07	Booking Cancellation	Valid ID: TKT-89K2M4	Seats freed in JSON, refund total confirmed.	PASS
TC-08	Non-Existent ID Cancel	ID: INVALID-99	System flags: [!] Booking ID not found .	PASS
TC-09	Persistence Verification	Restart application	Seats booked in previous session remain [X] .	PASS

11. Challenges Faced & Solutions

1. Challenge 1: Formatting Inconsistencies with String Prices

- *Problem:* Initial movie prices entered as text strings caused `ValueError: Unknown format code 'f' for object of type 'str'`.
- *Solution:* Sanitized the schema so ticket prices are stored as native floating-point numbers (`float`), ensuring seamless arithmetic and decimal formatting.

2. Challenge 2: Preventing Accidental Directory Name Clashes

- *Problem:* Creating a folder with the database name (`cinema_data.json`) led to OS permission errors (`PermissionError: [Errno 13]`).
- *Solution:* Structured paths cleanly and implemented exception guards inside `load_data()` to handle directory and file permission anomalies.

3. Challenge 3: Indentation Errors During Module Refactoring

- *Problem:* Moving logic into modular functions caused Python indentation mismatch errors (`IndentationError: unindent does not match any outer indentation level`).
- *Solution:* Standardized indentation strictly to 4 spaces across all modules and used automated VS Code document formatting (`Shift + Alt + F`).

12. Learnings & Key Takeaways

- **Modular Software Design:** Gained practical experience in architecting multi-module Python projects (`import storage` , `import display` , etc.) to keep codebases maintainable.
- **Defensive Programming:** Implemented input validation and sanitization at every step to prevent runtime crashes from unexpected user inputs.
- **Data Persistence:** Mastered JSON file input/output in Python to bridge the gap between volatile memory and durable disk storage.
- **Git & Version Control:** Learned best practices for structuring GitHub repositories, authoring descriptive documentation, and setting up clean project workflows.

13. Future Enhancements

- **Multi-Screen Scheduling:** Supporting multiple showtimes (Morning, Matinee, Evening) per movie per day.
- **Relational Database Migration:** Transitioning storage from JSON to SQLite or PostgreSQL using an ORM like SQLAlchemy.
- **Discount & Promo Codes:** Adding dynamic promotional code validation (e.g., `STUDENT10` for 10% off) during billing.
- **Role-Based Access Control:** Adding an Administrator login menu to add, edit, or delete screening films dynamically.

14. References

1. Python Software Foundation. *Python 3.12 Documentation - Built-in Types and Standard Library*. Available at: <https://docs.python.org/3/> 
2. Python `json` Module Documentation. Available at: <https://docs.python.org/3/library/json.html> 
3. GitHub Guides. *Mastering Markdown*. Available at: <https://guides.github.com/features/mastering-markdown/> 
4. Course Reference Material: *Python Essentials*, VITyarthi Learning Portal.